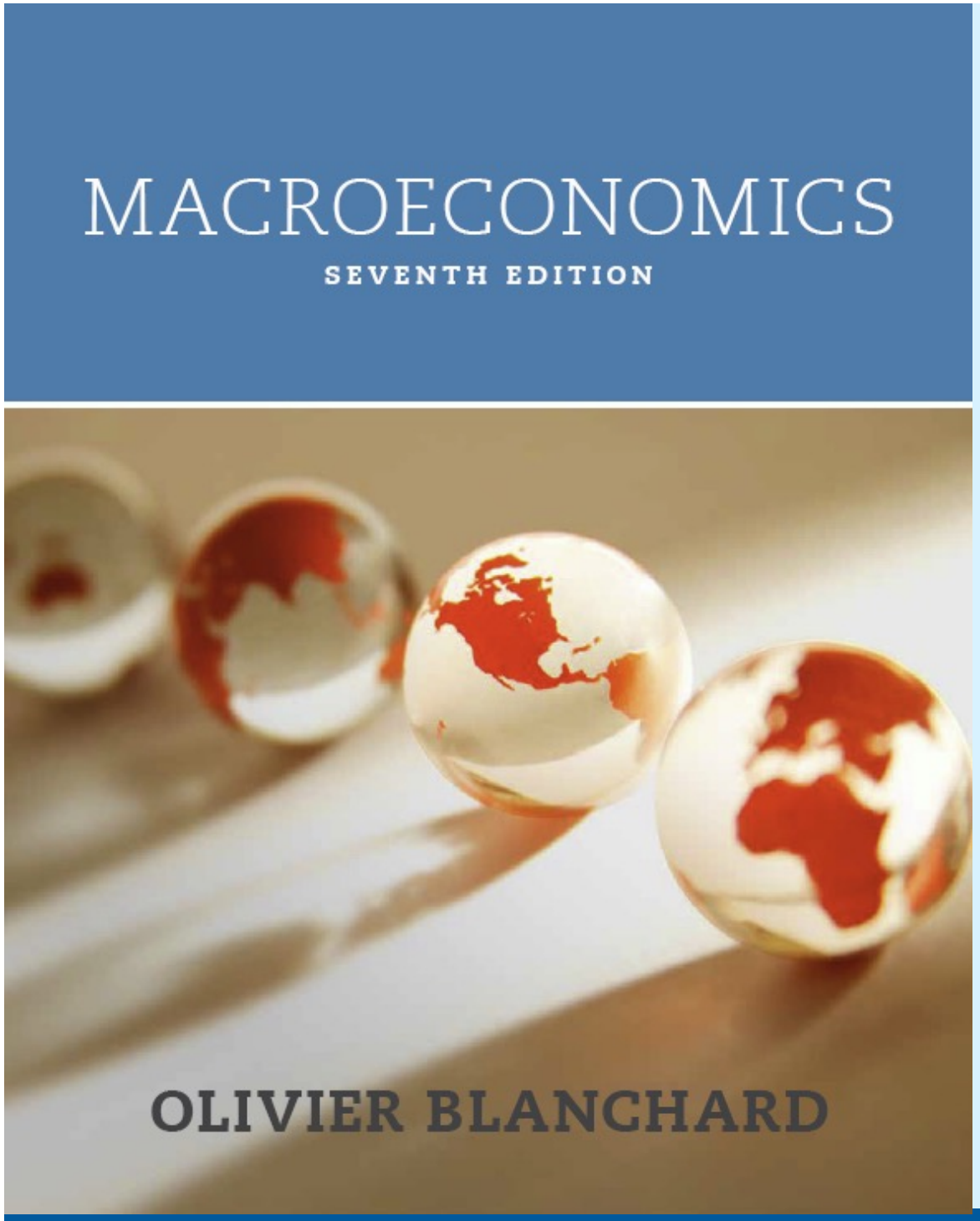


Financial Markets I

Chapter 4

ALWAYS LEARNING
PEARSON

Inflation in Spain soared to 4.9% in September, driven by rising fuel prices

Preliminary data suggest highest CPI in three and a half years



Several fuel hoses at a gas station / Geneviève Engel - Unió Europea

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Eurozone inflation

Eurozone inflation hits record 10% as energy prices continue to soar

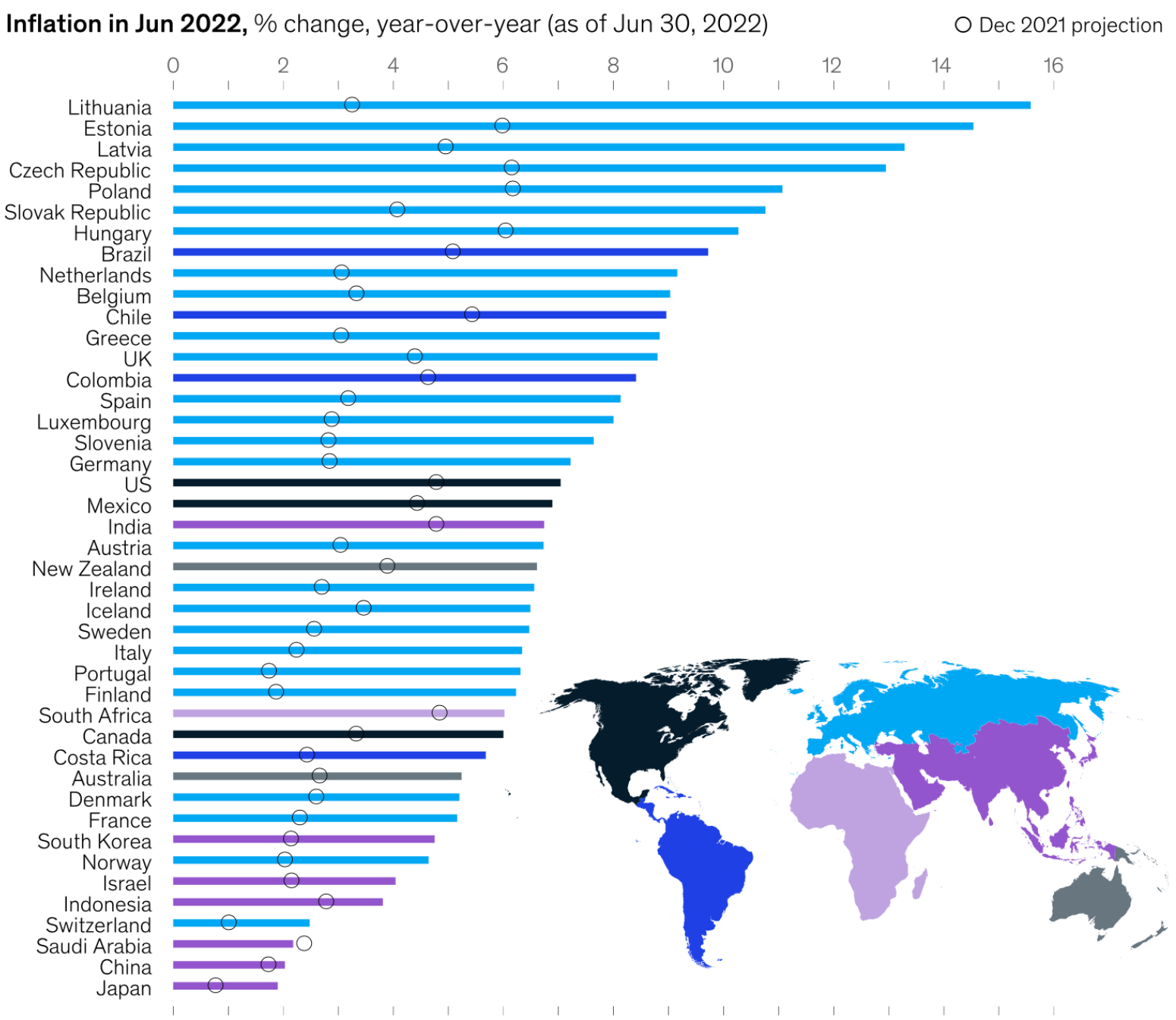
September 2022

September's double-digit figure bolsters calls for more big rate rises from ECB to slow price growth



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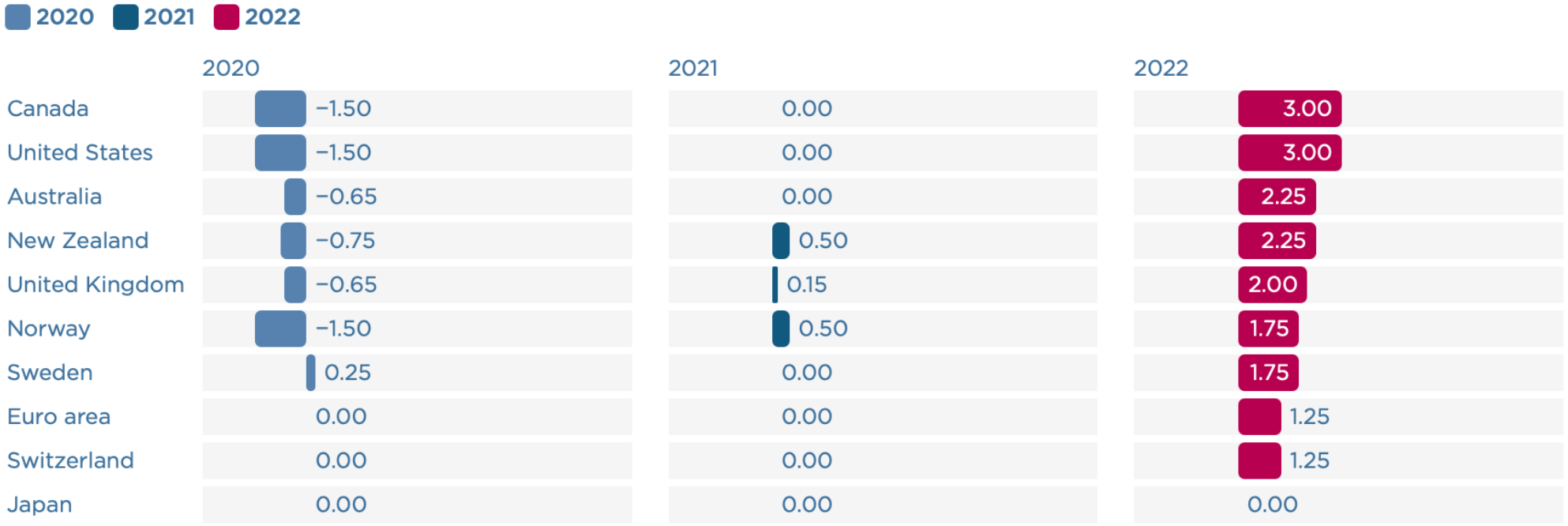
https://www.mckinsey.com/featured-insights/inflation/how-inflation-is-flipping-the-economic-script

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Central banks everywhere are raising rates, amplifying each other's policy impact

Year-on-year percentage point change in policy interest rate, as of September 27, 2022

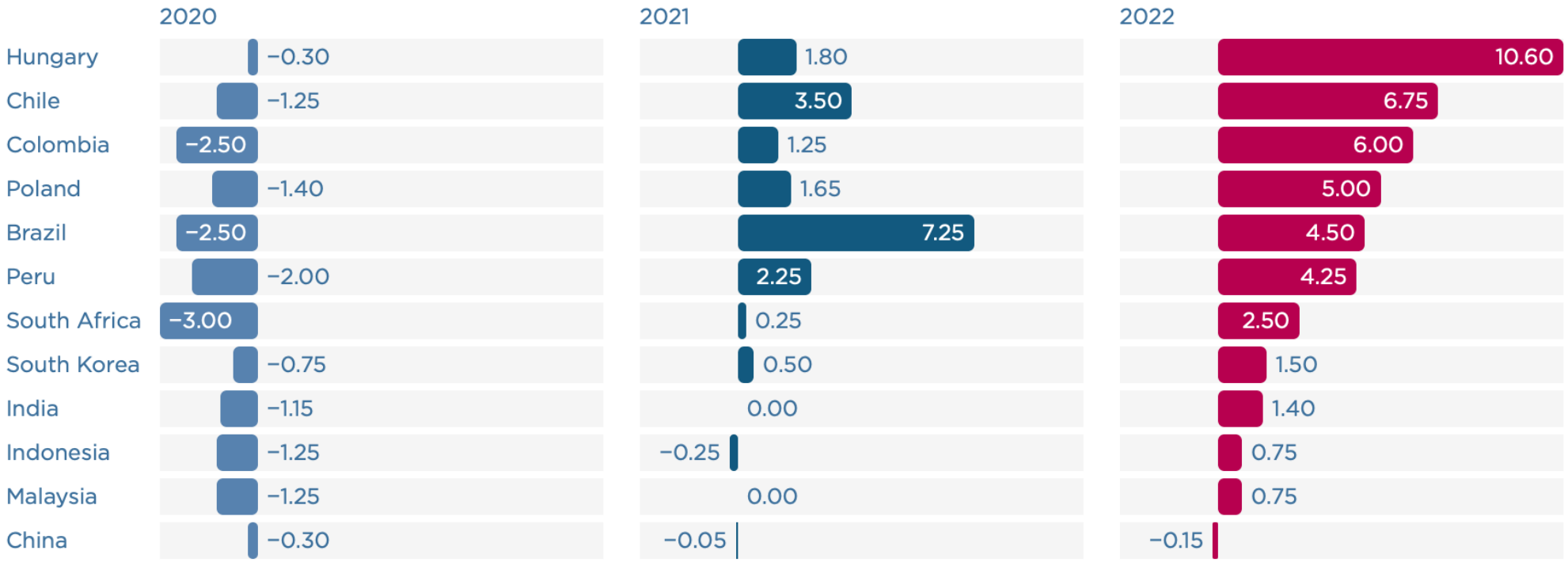
a. Advanced economies



https://www.piie.com/blogs/realtime-economic-issues-watch/uncoordinated-monetary-policies-risk-historic-global-slowdown?utm_source=update-newsletter&utm_medium=email&utm_campaign=piie-insider&utm_term=yyyy-mm-dd

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b. Emerging-market and developing economies



PIIE

Source: Central bank sources and Maurice Obstfeld and Haonan Zhou, "The Global Dollar Cycle," paper presented at the Brookings Papers on Economic Activity (BPEA) Conference, September 8-9, 2022.

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Consumer Price Index for All Urban Consumers (CPI-U)

12-Month Percent Change

Series Id: CUUR0000SA0L1E

Not Seasonally Adjusted

Series Title: All items less food and energy in U.S. city average, all urban consumers, not seasonally adjusted

Area: U.S. city average

Item: All items less food and energy

Base Period: 1982-84=100



Source: Bureau of Labor Statistics

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The Federal Reserve raised in September 2026
its benchmark interest rate by 25 basis points
to a target range of 3.75%-4%, its first increase since 2023.



https://fred.stlouisfed.org/series/DFF

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What do Traders Expect?



https://www.cmegroup.com/markets/interest-rates/cme-fedwatch-tool.html

https://edition.cnn.com/2023/09/20/economy/cme-fedwatch-tool-interest-rate-decision/index.html

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Outline

4-0	Background: Money & Central Banks
4-1	The Demand for Money
4-2	Determining the Interest Rate: I
4-3	Determining the Interest Rate: II
4-4	The Liquidity Trap

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Semantic Traps: Money, Income, Wealth

Stocks vs. Flows

- **Money** is what can be used to pay for transactions.
- **Income** is what you earn, and it is a flow.
- **Saving** is the part of after-tax income that you do not spend, and it is also a flow.
- **Savings** is the value of what you have accumulated over time.
- **Financial wealth**, or wealth, is the value of all your financial assets minus all your financial liabilities, and it is a stock variable
- **Investment** is what economists refer to as the purchase of new capital goods.
- **Financial investment** is the purchase of shares or other financial assets.

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4-0 Background. Money

Money is the stock of assets
that can be readily used to make transactions.

Fiat money

An asset that is used as legal tender by government decree
and is not backed by a physical commodity like gold.



European Central Bank MONEY DEFINITION

Table 2.1 Definitions of euro area monetary aggregates

Liabilities[1]	M1	M2	M3
Currency in circulation	x	x	x
Overnight deposits	x	x	x
Deposits with an agreed maturity of up to 2 years		x	x
Deposits redeemable at notice of up to 3 months		x	x
Repurchase agreements			x
Money market fund shares/units			x
Debt securities issued with a maturity of up to 2 years			x

Source: ECB.

1) Monetary liabilities of MFIs and central government (post office, treasury) vis-à-vis non-MFI euro area residents excluding central government.

M1: a narrow monetary aggregate that comprises currency in circulation plus overnight deposits held with MFIs and central government (e.g. at the post offi ce or treasury).

M2: an intermediate monetary aggregate that comprises M1 plus deposits redeemable at a period of notice of up to and including three months (i.e. short-term savings deposits) and deposits with an agreed maturity of up to and including two years (i.e. short-term time deposits) held with MFIs and central government.

M3: a broad monetary aggregate that comprises M2 plus marketable instruments, in particular repurchase agreements, money market fund shares and units, and debt securities with a maturity of up to and including two years issued by MFIs.

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Examples

- A student has €40 in her wallet and €600 in her current account. The total €640 is included in M1 because she can spend it immediately.

- A student has: €100 in cash; €900 in his current account; €3,000 in a one-year bank deposit. The first €1,000 is included in M1. The bank deposit is not included in M1 because it is not immediately available, but it is included in M2.

- A student has: €500 in her current account; €2,000 in a one-year bank deposit; €1,500 invested in a money market fund. M3

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4-0 Background. Money Supply

-The quantity of money available in the economy is the **Money Supply**: **M**.

-The Government controls the Money Supply.

- Control of the money supply is: **Monetary policy**

- Monetary policy is conducted by a country's **Central Bank**.

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4-1 The Demand for Money

Suppose you only have a choice between two assets:

Money and Bonds.



- *Money* is used for transactions, but it pays no interest.
- Two types of money: **Currency** and **Checkable deposits.**
- **Bonds** pay a positive interest rate, *i* (*the* rate of interest), but cannot be used for transaction.

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4-1 The Demand for Money

- The holding of money and bonds depends on:
  - Your level of transactions
  - The interest rate on bonds

- You can hold bonds indirectly through **money market funds:** is an open-ended mutual fund that invests in short-term debt securities such as US Treasury bills and commercial paper.

- In the early 1980s, the interest rate on money market funds reached 14% per year, so people earned more interest by moving their wealth from checking accounts to these funds.

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4-1 The Demand for Money

- Demand for money (M^d) is equal to nominal income $\$Y$ (a measure of level of transactions in the economy) times a decreasing function of the interest rate i:

$$M^d = \$Y\ L(i) \qquad (4.1)$$

$$(-)$$

- An increase in the interest rate decreases the demand for money, as people put more of their wealth into bonds.

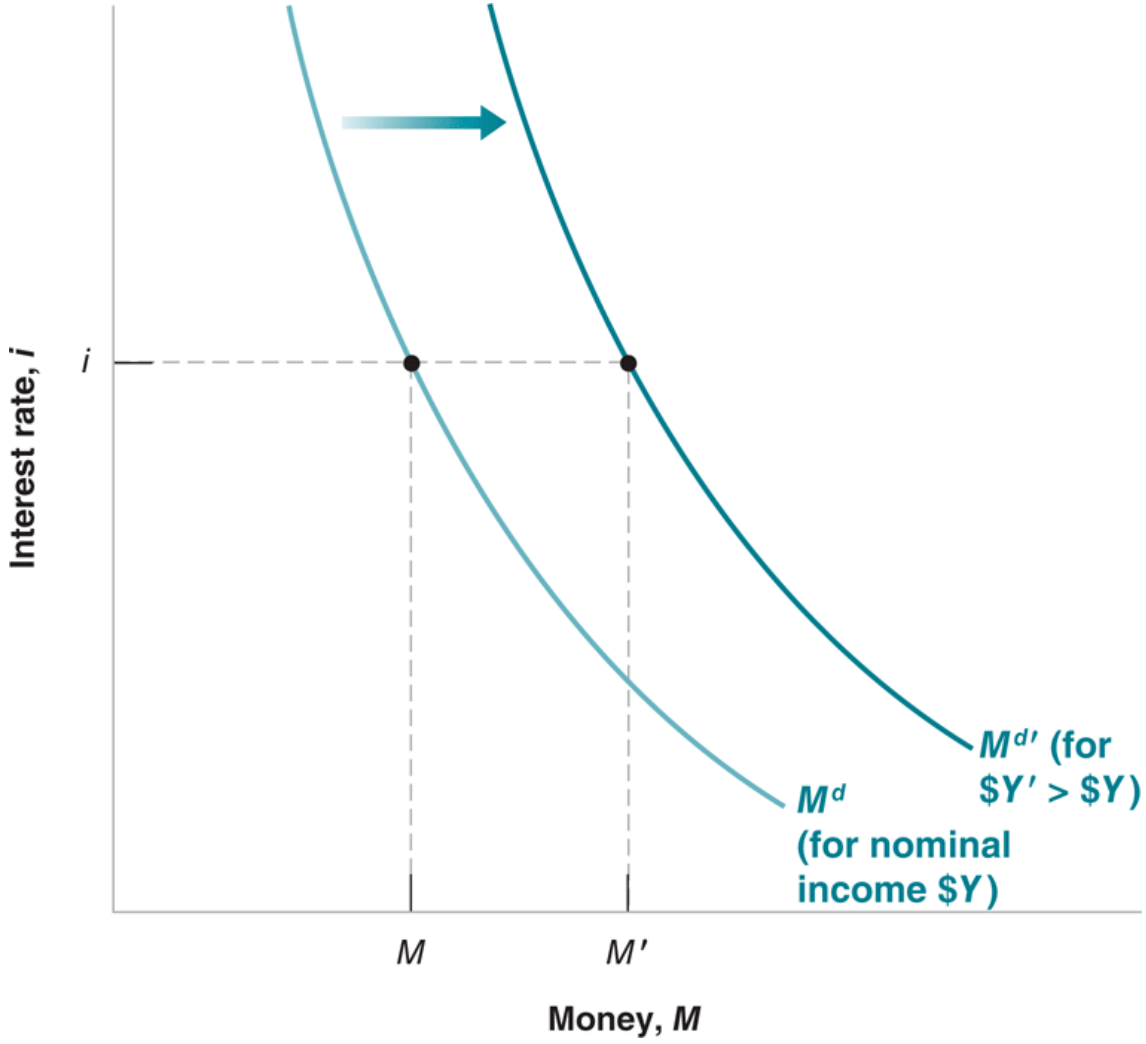
https://www.youtube.com/watch?v=Zt1Esq-qOZM [8.14 min]

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4-1 The Demand for Money

- **For a given level of nominal income**, a lower interest rate increases the demand for money.
- **At a given interest rate**, an increase in nominal income shifts the demand for money to the right.

Figure 4-1 The Demand for Money



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4-2 Determining the Interest Rate: I

- Suppose the central bank decides to supply an amount of money equal to M:

$$M^s = M$$

- Equilibrium in financial markets requires that $M^s = M^d = M$:

$$\text{Money supply} = \text{Money demand}$$

$$M = \$Y\ L(i) \qquad (4.2)$$

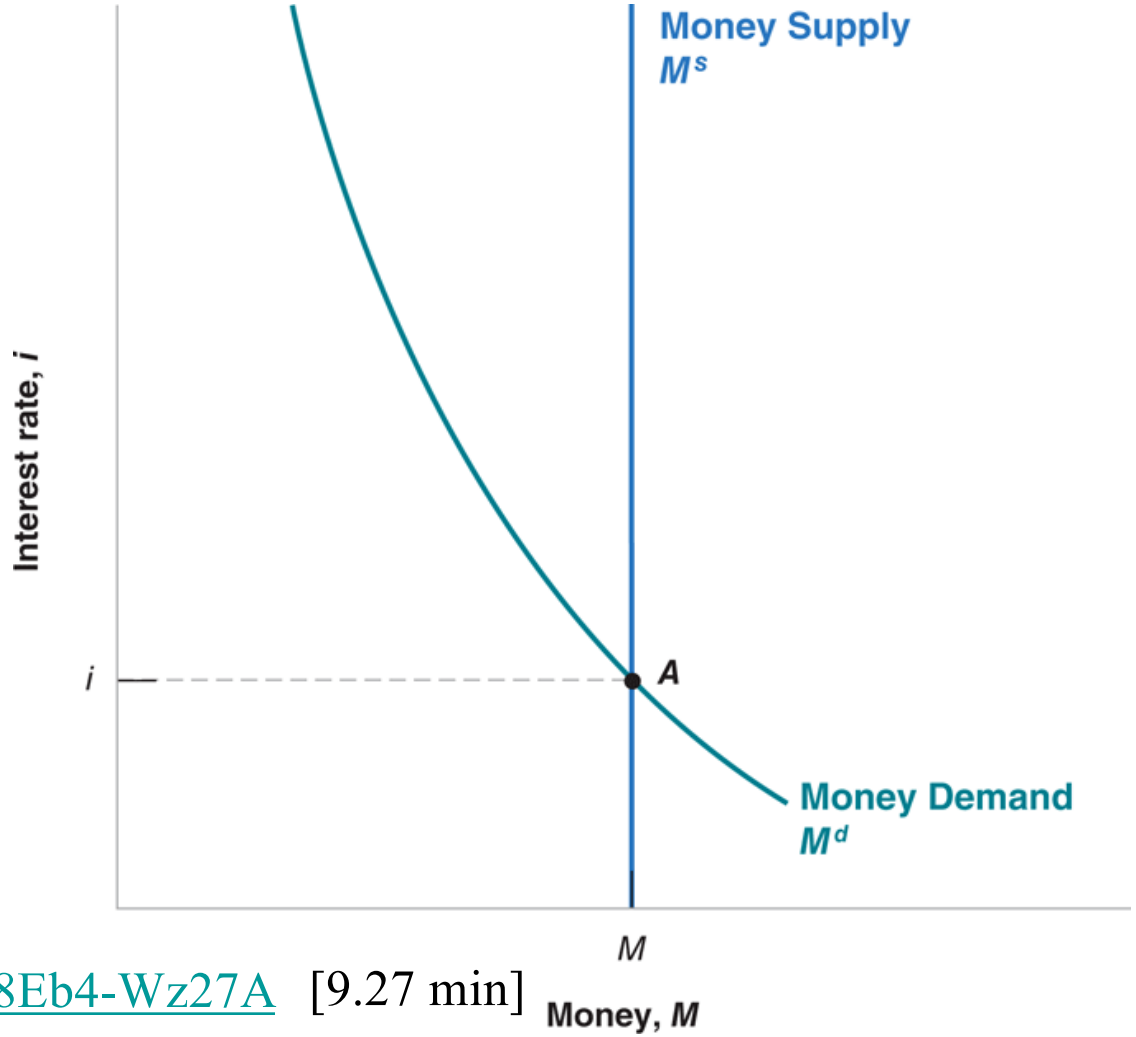
Assumption: Only CURRENCY

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4-2 Determining the Interest Rate: I

Figure 4-2 The Determination of the Interest Rate

The interest rate must be such that the supply of money (which is independent of the interest rate) is equal to the demand for money (which does depend on the interest rate).



https://www.youtube.com/watch?v=C8Eb4-Wz27A [9.27 min]

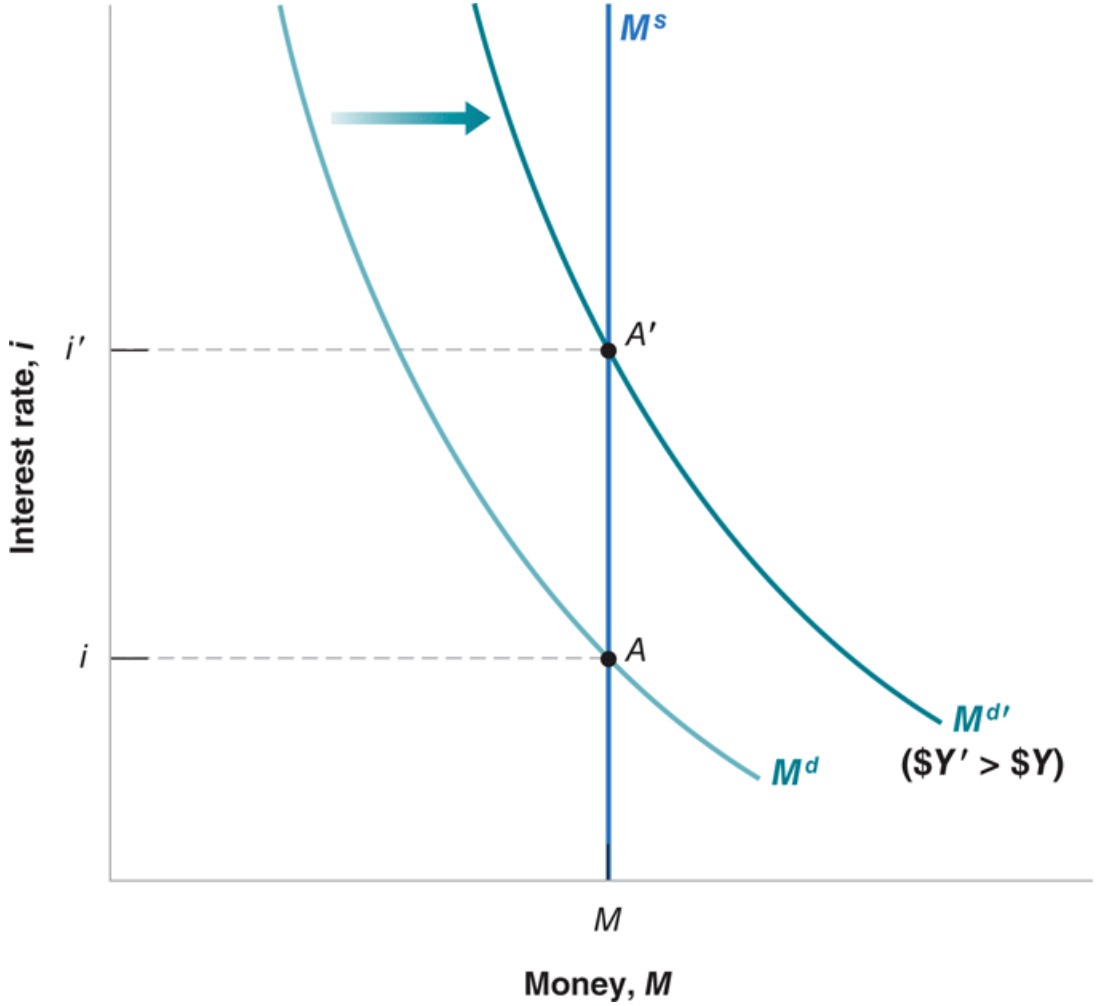
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4-2 Determining the Interest Rate: I

Figure 4-3 The Effects of an Increase in Nominal Income on the Interest Rate

Given the money supply, **an increase in nominal income** leads to an increase in the interest rate.

If $Y increases, people want to hold more money, Money Demand shifts.

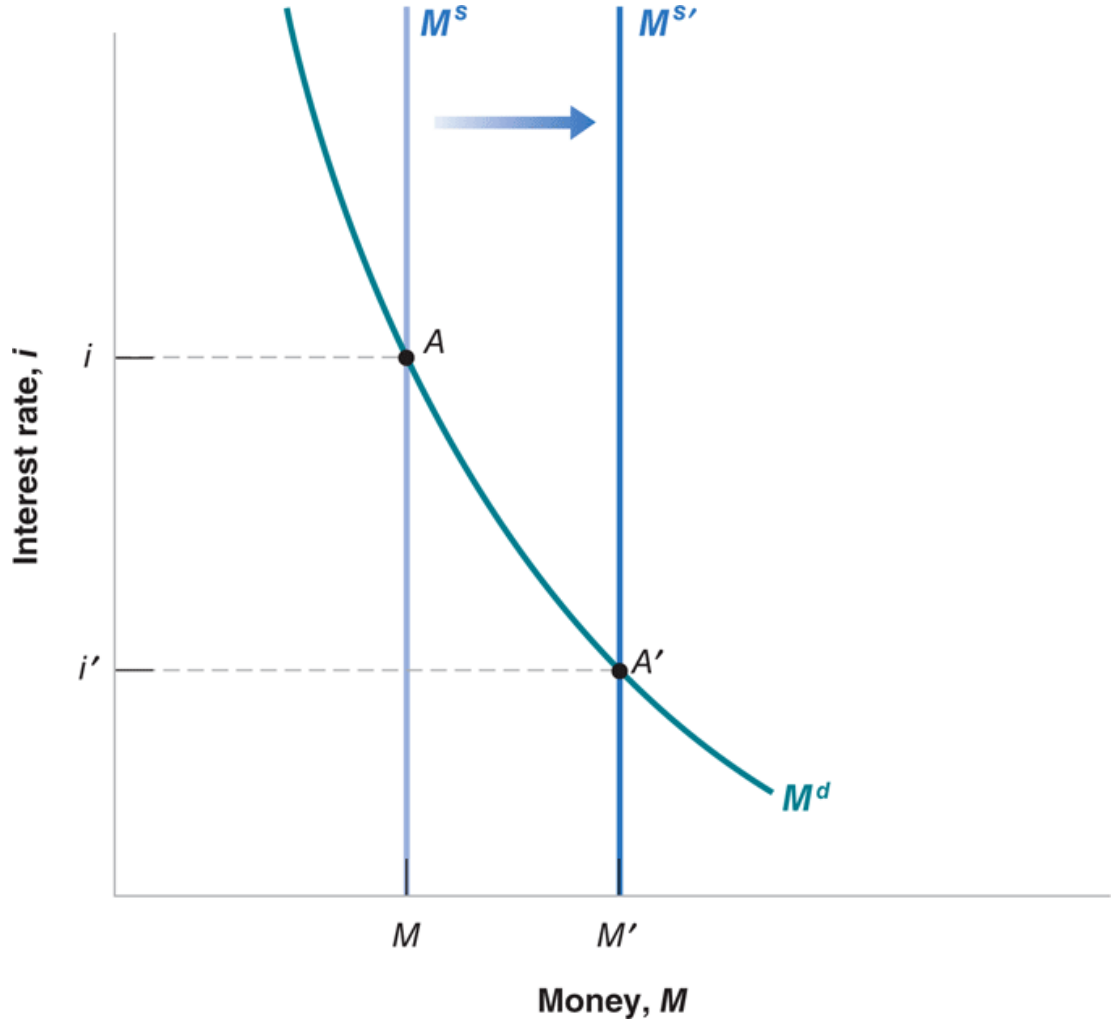


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4-2 Determining the Interest Rate: I

Figure 4-4 The Effects of an Increase in the Money Supply on the Interest Rate

An increase in the **supply of money** leads to a decrease in the interest rate.



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4-2 Determining the Interest Rate: I

- Central banks typically change the supply of money by buying or selling bonds in the bond market—**open market operations**.

- **Expansionary open market operation**: the central bank *expands* the supply of money by buying bonds.

- **Contractionary open market operation**: the central bank *contracts* the supply of money by selling bonds.

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4-2 Determining the Interest Rate: I

Figure 4-5 The Balance Sheet of the Central Bank and the Effects of an Expansionary Open Market Operation

- The assets of the central bank are the bonds it holds.
- The liabilities are the stock of money in the economy.

Balance Sheet

Assets	Liabilities
Bonds	Money (currency)

An open market operation in which the **central bank buys bonds** and issues **money (M^s rises)** increases both assets and liabilities by the same amount.

The Effects of an Expansionary Open Market Operation

Assets	Liabilities
Change in bond holdings: +$1 million	Change in money stock: +$1 million

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4-2 Determining the Interest Rate: I

- Suppose a bond such as a **Treasury bill**, or **T-bill,** promises to pay $100 a year from now.
- If the price of the bond today is $\$P_B$, then the interest rate on the bond is:

$$i = \frac{\$100 - \$P_B}{\$P_B}$$

- *The higher the price of the bond, the lower the interest rate.*
- *The higher the interest rate, the lower the price today.*

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4-2 Determining the Interest Rate: I

Expansionary Open Market Operation:

- Central Bank buys bonds: demand of bonds rises.
- Prices of bonds rise
- Interests decrease
- Money Supply increases.

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4-2 Determining the Interest Rate: I

- Rather than the money supply, the central bank could have **chosen the interest rate** and then adjusted the money supply so as to achieve the interest rate it had chosen.

- Choosing the interest rate, instead of the money supply, is what modern central banks, including the Fed, typically do.

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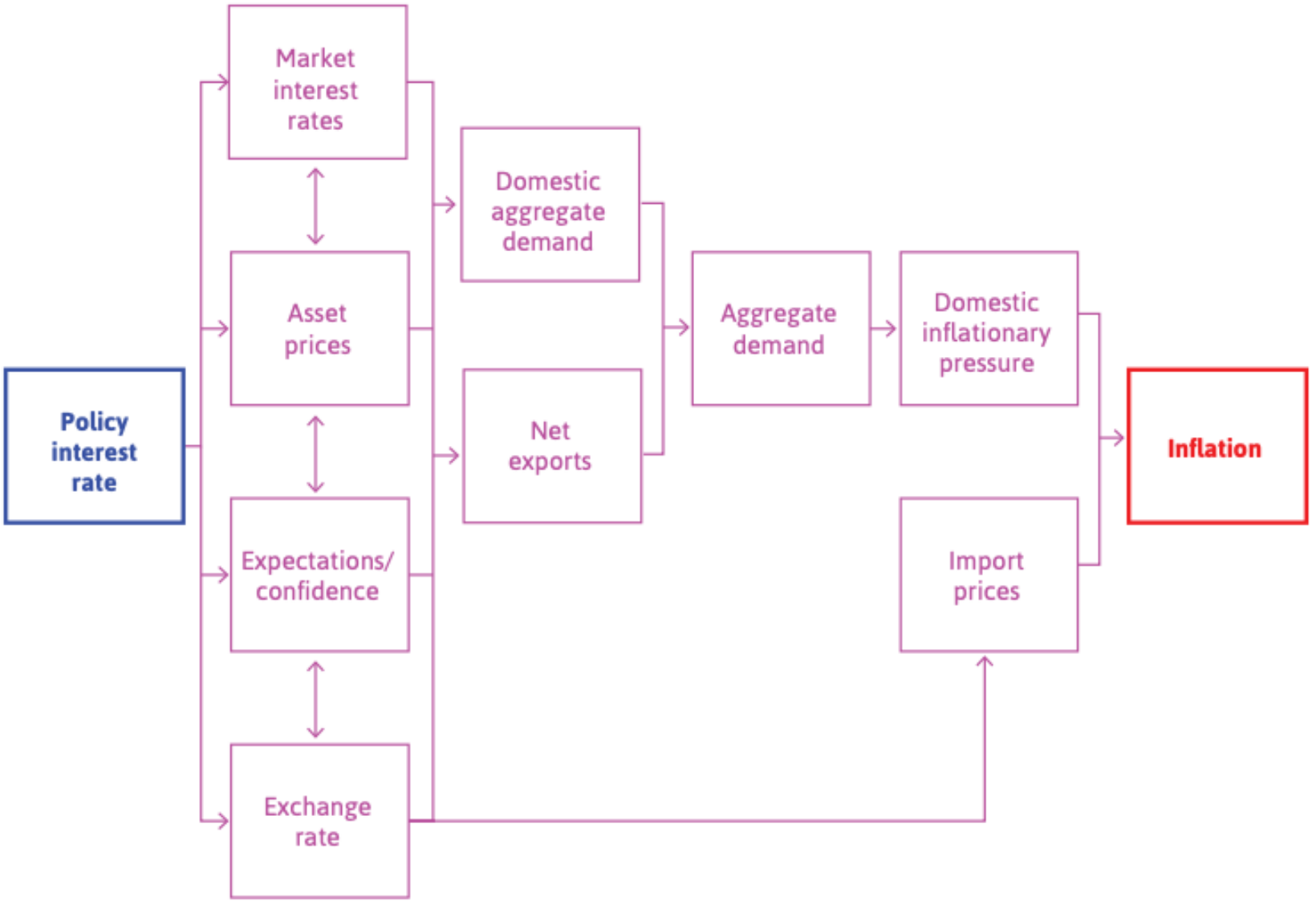


Figure 15.14 Monetary policy transmission mechanisms.

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Open market operations (OMOs) and **setting the federal funds rate** are both essential tools used by the U.S. Federal Reserve to influence monetary policy. The relationship between them is key to how monetary policy is implemented.

1. Federal Funds Rate as the Target

•The **federal funds rate** is the primary tool or *target* the Federal Reserve uses to influence the broader economy. The Federal Open Market Committee (FOMC) meets regularly to set a target range for the federal funds rate, which is the interest rate at which banks lend to each other overnight.

•The **federal funds** rate refers to the interest rate at which banks and other depository institutions lend excess reserves to each other overnight.

2. Open Market Operations as the Tool to Achieve the Target

•While the Fed **sets a target** for the federal funds rate, it doesn't directly dictate that rate. Instead, the Fed uses **open market operations** to adjust the supply of reserves in the banking system, thereby guiding the actual federal funds rate toward the desired target.

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- Since the Great Recession, and during the large-scale asset purchases (Quantitative Easing), central banks built very large balance sheets.
- Reserves in the banking system became *abundant*, far above the levels banks would hold for transaction/liquidity needs alone.
- In this situation the classical money supply = money demand logic is no longer a useful operational tool for controlling interest rates.
- Small changes in reserve supply don't have an effect on the interbank rate (because the supply is already so large that bank demand is relatively inelastic).
- Central banks move to a **floor** or **so-called "ample reserves"** system: **they pay interest on reserves.**
- The **interest on reserves (IOR)** becomes a key tool.

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- If excess reserves are abundant, then banks will not lend reserves at a rate below the IOR, because they could just leave them at the central bank and get that rate. Thus the IOR acts as a *floor* to the interbank rate (or a closely related benchmark).

- In addition, central banks often provide standing lending/borrowing facilities (repos, reverse repos, etc.) to absorb or inject liquidity and manage extreme fluctuations.

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Comparison: Fed vs ECB Policy Rates Both operate in a 'floor system'

🇺🇸 Federal Reserve (U.S.)

- Policy rate: Federal Funds Rate (target range)
- Interest on reserves: Interest on Reserve Balances (IORB)

🇪🇺 European Central Bank (ECB)

- Policy rate: Deposit Facility Rate (DFR)
- Interest on reserves: Deposit Facility Rate

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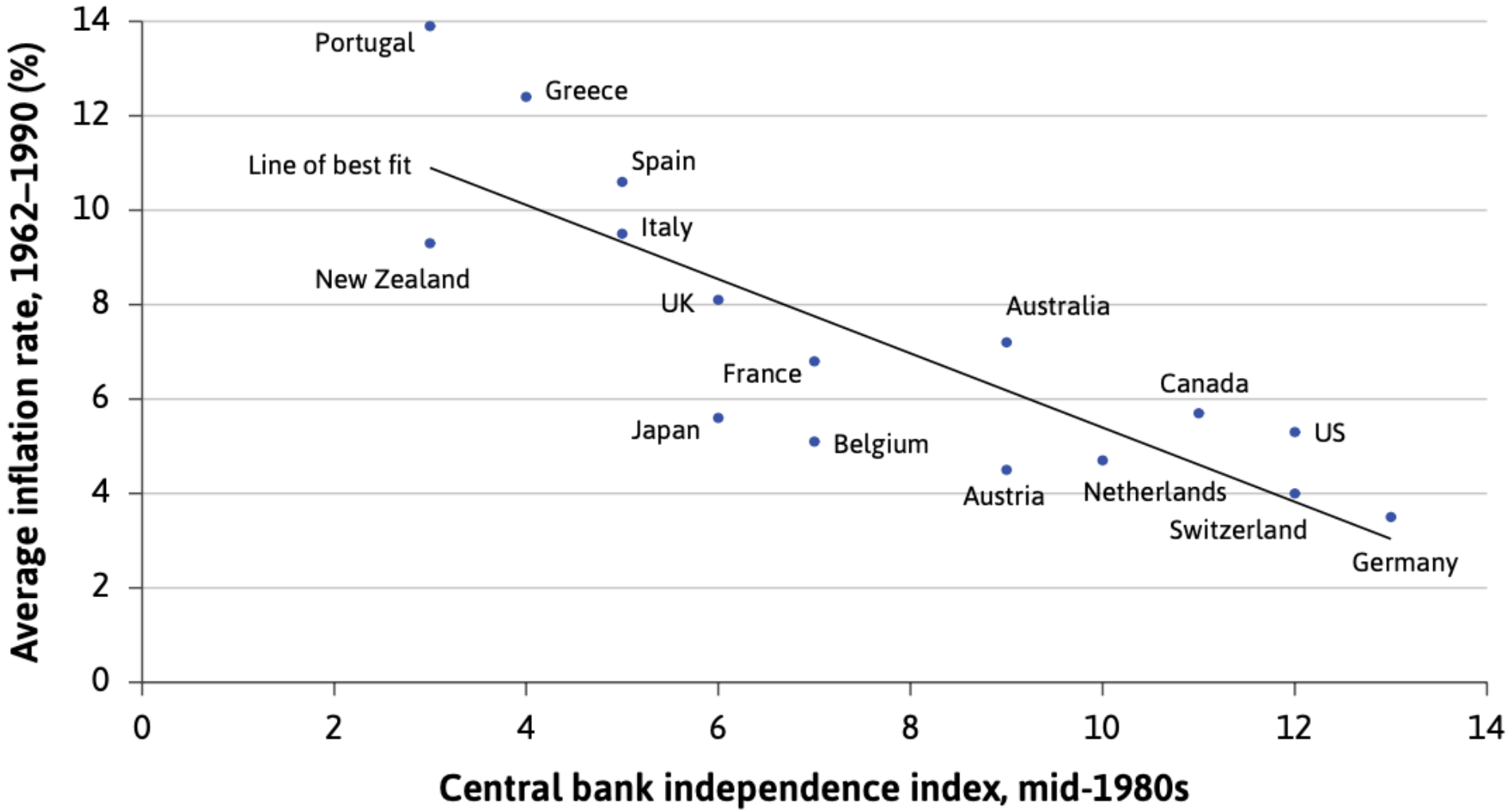


Figure 15.20 Inflation and central bank independence: OECD countries.

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4-3 Determining the Interest Rate: II

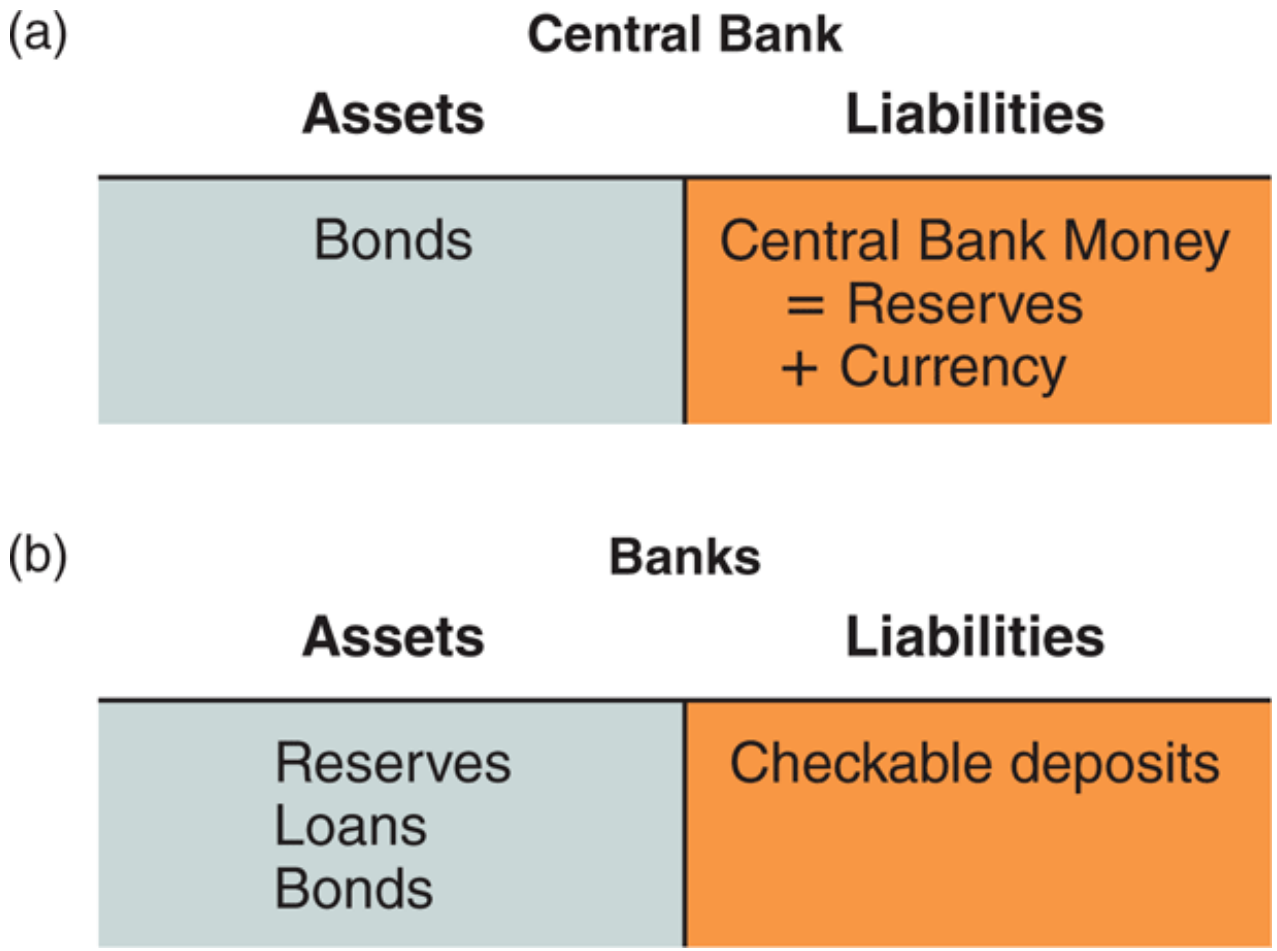
- **Financial intermediaries**: Institutions that receive funds from people and firms and use these funds to buy financial assets or to make loans to other people and firms.
- Banks are financial intermediaries that have money, in the form of *checkable deposits,* as their liabilities.
- Banks keep as **reserves** some of the funds they receive.
- The liabilities of the central bank are the money it has issued, called **central bank money**.

Assumption: CURRENCY+ RESERVES

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4-3 Determining the Interest Rate: II

Figure 4-6 The Balance Sheet of Banks, and the Balance Sheet of the Central Bank Revisited



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4-3 Determining the Interest Rate: II

- Assume people hold no currency so the demand for money by people is the demand for checkable deposits:

$$M^d = \$Y\ L(i) \qquad (4.3)$$
$$(-)$$

- The demand for reserves by banks depends on the amount of checkable deposits:

$$H^d = \theta M^d = \theta \$Y\ L(i) \qquad (4.4)$$

- θ is the reserve ratio, and H^d is demand for high-power money or the monetary base.

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4-3 Determining the Interest Rate: II

- Let H denote the supply of central bank money, then the equilibrium condition:

$$H = H^d \tag{4.5}$$

- Or using equation (4.4):

$$H = \theta \$Y \; L(i) \tag{4.6}$$

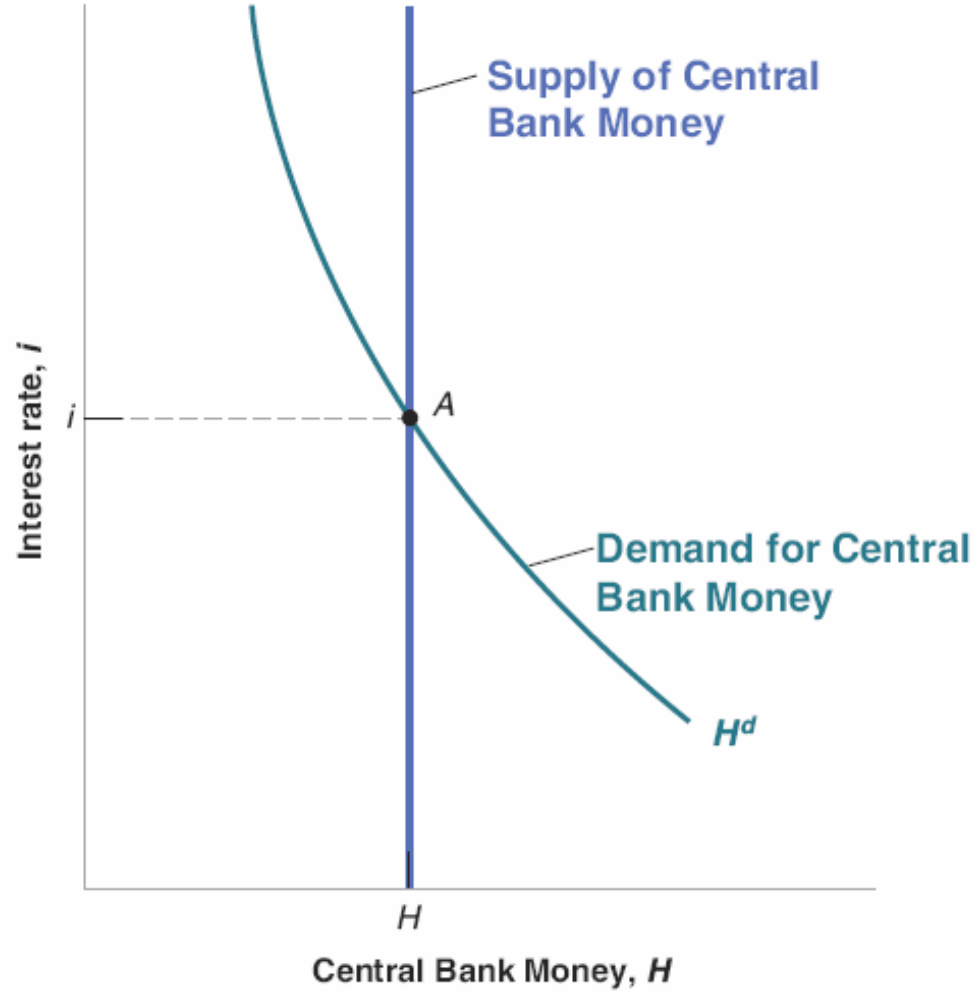
- An increase in H leads to a decrease in the interest rate, and a decrease in H leads to an increase in the interest rate.

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4-3 Determining the Interest Rate: II

Figure 4-8 Equilibrium in the Market for Central Bank Money and the Determination of the Interest Rate

The equilibrium interest rate is such that the supply of central bank money is equal to the demand for central bank money.



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4-4 The Liquidity Trap

- **Zero lower bound**: The interest rate cannot go below zero.

- The economy is in a **liquidity trap** when the interest rate is down to zero, monetary policy cannot decrease it further.

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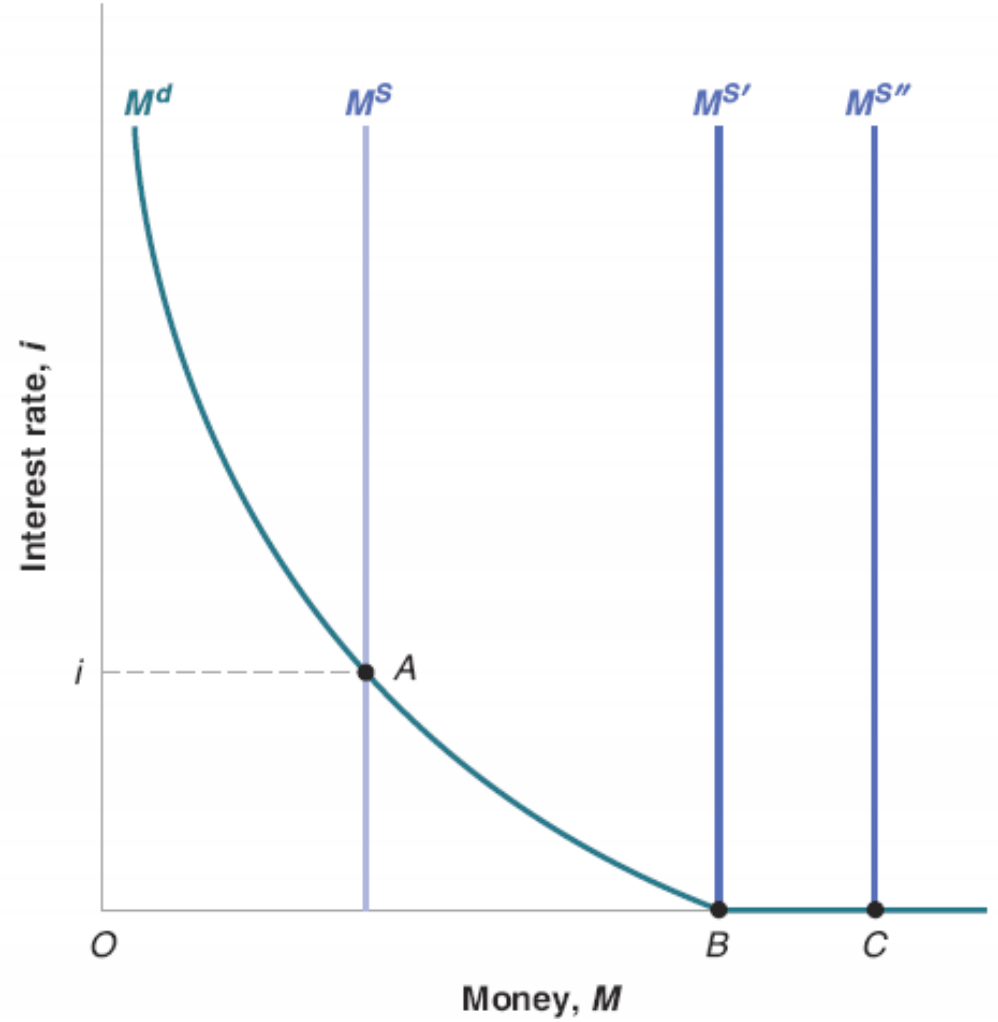
4-4 The Liquidity Trap

Figure 4-8 Money Demand, Money Supply, and the Liquidity Trap

When the interest rate is equal to zero, and once people have enough money for transaction purposes, they become indifferent between holding money and holding bonds.

The demand for money becomes horizontal.

This implies that, when the interest rate is equal to zero, further increases in the money supply have no effect on the interest rate, which remains equal to zero.

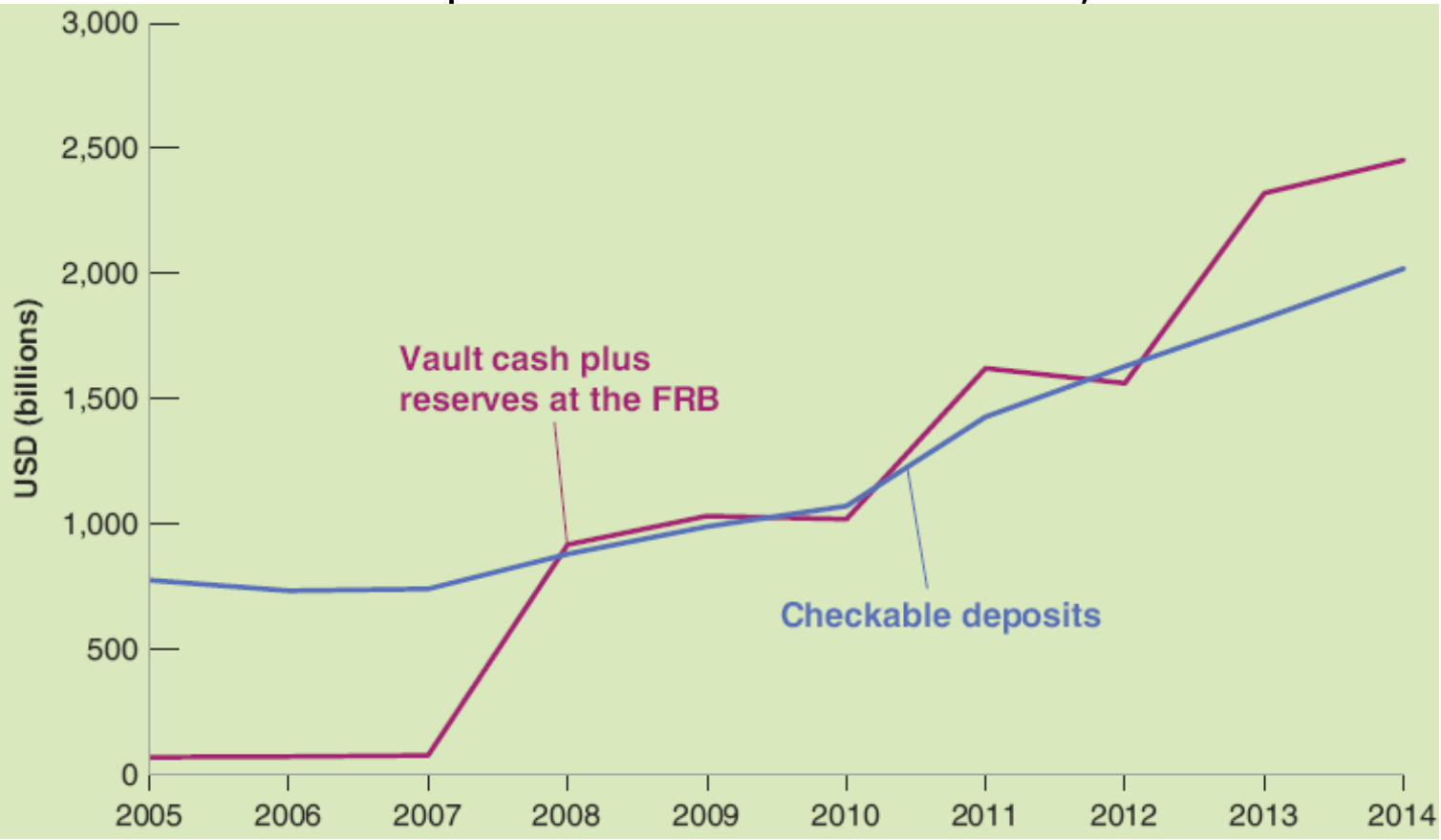


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FOCUS: The Liquidity Trap in Action

- The large increase in the supply of central bank money between 2007 and 2014 was absorbed by households and banks.

Figure 1 Checkable Deposits and Bank Reserves, 2005–2014.



Source: Flow of Funds.

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CASE STUDY: Monetary contraction and interest rates

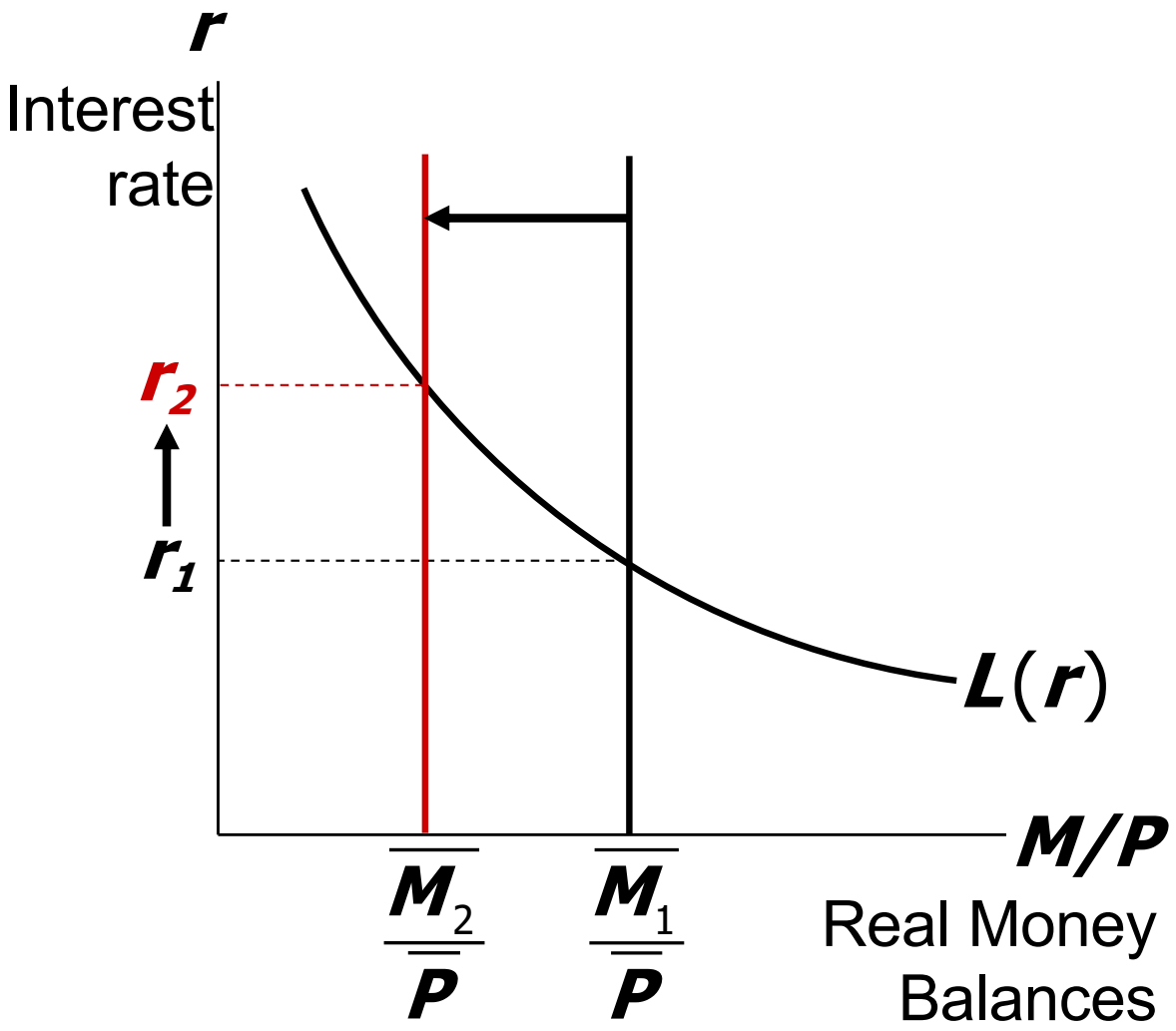
- Late 70s in U.S.: $\pi > 10\%$
- Oct 1979: The president from the Fed, Paul Volcker announces that the goal of monetary policy is to reduce inflation
- August 1979-April 1980: The Fed reduces ***M/P*** 8,0%
- January 1983: $\pi = 3{,}7\%$

How did this policy change affect nominal interest rates?

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How does the Central Bank increase interest rates?

To raise ***r***, the Central Bank reduces ***M***



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